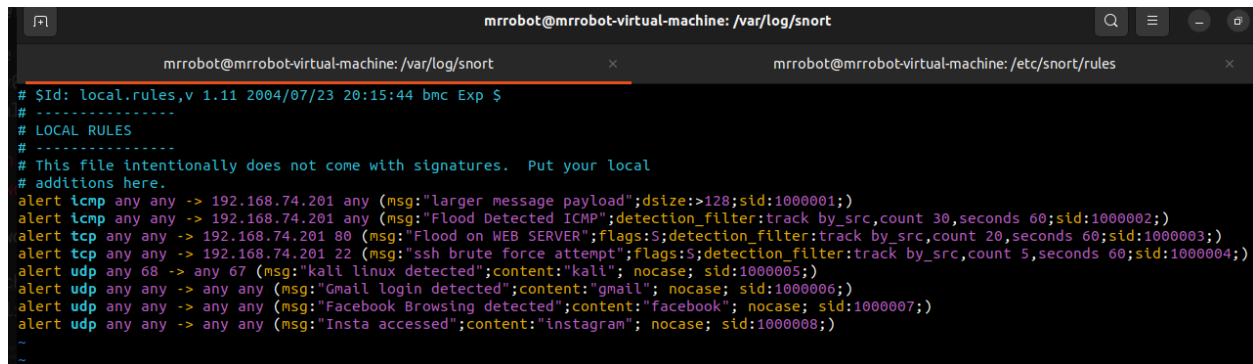


ASSIGNMENT 3

Install a web server on a Linux system. Install and configure Snort as IDS on this system. Add following rules to Snort to generate an alert for following threats.

1. If a ping packet with size greater than 128 bytes is received.
2. If from an outside IP address 30 ICMP packets are received within 1 min.
3. If from an outside single IP address more that 20 TCP SYN requests are received in 1 min to the web server
4. If a brute force attack is done on the SSH service.
5. If a kali Linux machine,boots in the network.
6. If a gmail website is accessed from the web server.
7. If the facebook website is accessed from the web server.
8. If the Instagram website is accessed from the web server.



```
mrrobot@mrrobot-virtual-machine: /var/log/snort
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
#
# LOCAL RULES
#
# This file intentionally does not come with signatures. Put your local
# additions here.
alert icmp any any -> 192.168.74.201 any (msg:"larger message payload";dsize:>128;sid:1000001;)
alert icmp any any -> 192.168.74.201 any (msg:"Flood Detected ICMP";detection_filter:track by_src,count 30,seconds 60;sid:1000002;)
alert tcp any any -> 192.168.74.201 80 (msg:"Flood on WEB SERVER";flags:S;detection_filter:track by_src,count 20,seconds 60;sid:1000003;)
alert tcp any any -> 192.168.74.201 22 (msg:"ssh brute force attempt";flags:S;detection_filter:track by_src,count 5,seconds 60;sid:1000004;)
alert udp any 68 -> any 67 (msg:"kali linux detected";content:"kali";nocase;sid:1000005;)
alert udp any any -> any any (msg:"Gmail login detected";content:"gmail";nocase;sid:1000006;)
alert udp any any -> any any (msg:"Facebook Browsing detected";content:"facebook";nocase;sid:1000007;)
alert udp any any -> any any (msg:"Insta accessed";content:"instagram";nocase;sid:1000008;)
```

```
C:\Users\PG-ITISS\Downloads>ping -l 500 192.168.74.201
```

```
Pinging 192.168.74.201 with 500 bytes of data:
```

```
Reply from 192.168.74.201: bytes=500 time<1ms TTL=64
```

```
Reply from 192.168.74.201: bytes=500 time<1ms TTL=64
```

```
Reply from 192.168.74.201: bytes=500 time<1ms TTL=64
```

```
Ping statistics for 192.168.74.201:
```

```
Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```

06/09-16:39:15.492711 [**] [1:1000001:0] larger message payload [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201
06/09-16:39:16.511118 [**] [1:1000001:0] larger message payload [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201
06/09-16:39:17.524655 [**] [1:1000001:0] larger message payload [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201
06/09-16:39:35.694913 [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPv6-ICMP} :: -> ff02::16
06/09-16:39:35.759117 [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPv6-ICMP} :: -> ff02::1:ff2e:3f93
06/09-16:39:36.367266 [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPv6-ICMP} :: -> ff02::16
06/09-16:40:43.433525 [**] [1:1000001:0] larger message payload [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201
06/09-16:40:44.447708 [**] [1:1000001:0] larger message payload [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201
06/09-16:40:45.459703 [**] [1:1000001:0] larger message payload [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201
06/09-16:40:46.468345 [**] [1:1000001:0] larger message payload [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201
mrrobot@mrrobot-virtual-machine: /var/log/snort$

```

```

mrrobot@mrrobot-virtual-machine: /var/log/snort$ cat snort.alert.fast | grep "larger"
06/09-15:33:58.032311 [**] [1:1000001:0] larger message payload [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201
06/09-15:33:59.038739 [**] [1:1000001:0] larger message payload [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201
06/09-15:34:00.057765 [**] [1:1000001:0] larger message payload [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201
06/09-15:34:01.071947 [**] [1:1000001:0] larger message payload [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201
06/09-16:39:15.492711 [**] [1:1000001:0] larger message payload [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201
06/09-16:39:16.511118 [**] [1:1000001:0] larger message payload [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201
06/09-16:39:17.524655 [**] [1:1000001:0] larger message payload [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201
06/09-16:40:43.433525 [**] [1:1000001:0] larger message payload [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201
06/09-16:40:44.447708 [**] [1:1000001:0] larger message payload [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201
06/09-16:40:45.459703 [**] [1:1000001:0] larger message payload [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201
06/09-16:40:46.468345 [**] [1:1000001:0] larger message payload [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201

```

```

mrrobot@mrrobot-virtual-machine: /var/log/snort$ cat snort.alert.fast | grep "Flood Detected ICMP"
06/09-16:49:42.184789 [**] [1:1000002:0] Flood Detected ICMP [**] [Priority: 0] {ICMP} 192.168.74.58 -> 192.168.74.201

```

```

(mrrobot@kali)-[~]
$ sudo hping3 -S -p 80 --flood 192.168.74.201
HPING 192.168.74.201 (eth0 192.168.74.201): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown

```

```

mrrobot@mrrobot-virtual-machine: /var/log/snort$ cat snort.alert.fast | grep "WEB SERVER" | head
06/09-16:48:54.648933 [**] [1:1000003:0] Flood on WEB SERVER [**] [Priority: 0] {TCP} 192.168.74.170:2843 -> 192.168.74.201:80
06/09-16:48:54.648952 [**] [1:1000003:0] Flood on WEB SERVER [**] [Priority: 0] {TCP} 192.168.74.170:2844 -> 192.168.74.201:80
06/09-16:48:54.649062 [**] [1:1000003:0] Flood on WEB SERVER [**] [Priority: 0] {TCP} 192.168.74.170:2970 -> 192.168.74.201:80
06/09-16:48:54.649081 [**] [1:1000003:0] Flood on WEB SERVER [**] [Priority: 0] {TCP} 192.168.74.170:2971 -> 192.168.74.201:80
06/09-16:48:54.649311 [**] [1:1000003:0] Flood on WEB SERVER [**] [Priority: 0] {TCP} 192.168.74.170:2972 -> 192.168.74.201:80
06/09-16:48:54.649332 [**] [1:1000003:0] Flood on WEB SERVER [**] [Priority: 0] {TCP} 192.168.74.170:2973 -> 192.168.74.201:80
06/09-16:48:54.649453 [**] [1:1000003:0] Flood on WEB SERVER [**] [Priority: 0] {TCP} 192.168.74.170:2974 -> 192.168.74.201:80
06/09-16:48:54.649473 [**] [1:1000003:0] Flood on WEB SERVER [**] [Priority: 0] {TCP} 192.168.74.170:2975 -> 192.168.74.201:80
06/09-16:48:54.649581 [**] [1:1000003:0] Flood on WEB SERVER [**] [Priority: 0] {TCP} 192.168.74.170:2976 -> 192.168.74.201:80
06/09-16:48:54.649601 [**] [1:1000003:0] Flood on WEB SERVER [**] [Priority: 0] {TCP} 192.168.74.170:2977 -> 192.168.74.201:80

```

```

(mrrobot@kali)-[~]
$ hydra -l user.txt -P pass.txt ssh://192.168.74.201
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in
military or secret service organizations, or for illegal purposes (this is n
on-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-09 11:
26:05
[WARNING] Many SSH configurations limit the number of parallel tasks, it is r
ecommended to reduce the tasks: use -t 4
[DATA] max 8 tasks per 1 server, overall 8 tasks, 8 login tries (l:1/p:8), ~1
try per task
[DATA] attacking ssh://192.168.74.201:22/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-06-09 11:26:11

```

```

mrrobot@mrrobot-virtual-machine:/var/log/snort$ cat snort.alert.fast | grep "brute force" | head
06/09-16:55:11.338071  [**] [1:1000004:0] ssh brute force attempt [**] [Priority: 0] {TCP} 192.168.74.170:37220 -> 192.168.74.201:22
06/09-16:55:11.338441  [**] [1:1000004:0] ssh brute force attempt [**] [Priority: 0] {TCP} 192.168.74.170:37226 -> 192.168.74.201:22
06/09-16:55:11.338953  [**] [1:1000004:0] ssh brute force attempt [**] [Priority: 0] {TCP} 192.168.74.170:37240 -> 192.168.74.201:22
06/09-16:55:11.339420  [**] [1:1000004:0] ssh brute force attempt [**] [Priority: 0] {TCP} 192.168.74.170:37248 -> 192.168.74.201:22

```

```

mrrobot@mrrobot-virtual-machine:/var/log/snort$ cat snort.alert.fast | grep "kali" | head
06/09-16:44:54.125203  [**] [1:1000005:0] kali linux detected [**] [Priority: 0] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
mrrobot@mrrobot-virtual-machine:/var/log/snort$

```

```

mrrobot@mrrobot-virtual-machine:/var/log/snort$ dig www.gmail.com

; <<>> DiG 9.18.39-0ubuntu0.22.04.4-Ubuntu <<>> www.gmail.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<- opcode: QUERY, status: NOERROR, id: 53504
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4000
;; QUESTION SECTION:
;www.gmail.com.                IN      A

;; ANSWER SECTION:
www.gmail.com.                 237     IN      A      142.250.76.165

;; Query time: 1 msec
;; SERVER: 192.168.72.20#53(192.168.72.20) (UDP)
;; WHEN: Tue Jun 09 17:23:05 IST 2026
;; MSG SIZE rcvd: 58

```

```

mrrobot@mrrobot-virtual-machine:/var/log/snort$ cat snort.alert.fast | grep "Gmail"
06/09-17:13:13.504523  [**] [1:1000006:0] Gmail login detected [**] [Priority: 0] {UDP} 192.168.74.58:57481 -> 192.168.72.20:53
06/09-17:13:13.504525  [**] [1:1000006:0] Gmail login detected [**] [Priority: 0] {UDP} 192.168.74.58:55807 -> 192.168.72.20:53
06/09-17:13:13.505305  [**] [1:1000006:0] Gmail login detected [**] [Priority: 0] {UDP} 192.168.74.58:61073 -> 192.168.72.20:53
06/09-17:13:13.505306  [**] [1:1000006:0] Gmail login detected [**] [Priority: 0] {UDP} 192.168.74.58:54986 -> 192.168.72.20:53
06/09-17:13:19.693370  [**] [1:1000006:0] Gmail login detected [**] [Priority: 0] {UDP} 192.168.74.58:62223 -> 192.168.72.20:53
06/09-17:13:19.693376  [**] [1:1000006:0] Gmail login detected [**] [Priority: 0] {UDP} 192.168.74.58:49705 -> 192.168.72.20:53
06/09-17:16:14.489468  [**] [1:1000006:0] Gmail login detected [**] [Priority: 0] {UDP} 192.168.74.58:65427 -> 192.168.72.20:53
06/09-17:16:14.489473  [**] [1:1000006:0] Gmail login detected [**] [Priority: 0] {UDP} 192.168.74.58:62994 -> 192.168.72.20:53
06/09-17:18:08.791700  [**] [1:1000006:0] Gmail login detected [**] [Priority: 0] {UDP} 192.168.72.20:53 -> 192.168.74.201:52049
06/09-17:23:03.176226  [**] [1:1000006:0] Gmail login detected [**] [Priority: 0] {UDP} 192.168.72.20:53 -> 192.168.74.201:45719
06/09-17:23:05.096762  [**] [1:1000006:0] Gmail login detected [**] [Priority: 0] {UDP} 192.168.72.20:53 -> 192.168.74.201:38891

```

```

mrrobot@mrrobot-virtual-machine:/var/log/snort$ dig www.facebook.com

; <<>> DiG 9.18.39-0ubuntu0.22.04.4-Ubuntu <<>> www.facebook.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 17459
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4000
;; QUESTION SECTION:
;www.facebook.com.                IN      A

;; ANSWER SECTION:
www.facebook.com.                1255    IN      CNAME   star-mini.c10r.facebook.com.
star-mini.c10r.facebook.com.    32      IN      A       57.144.42.1

;; Query time: 1 msec
;; SERVER: 192.168.72.20#53(192.168.72.20) (UDP)
;; WHEN: Tue Jun 09 17:24:29 IST 2026
;; MSG SIZE rcvd: 90

```

```

mrrobot@mrrobot-virtual-machine:/var/log/snort$ cat snort.alert.fast | grep "Facebook"
06/09-17:24:29.920405  [**] [1:1000007:0] Facebook Browsing detected [**] [Priority: 0] {UDP} 192.168.72.20:53 -> 192.168.74.201:60655
mrrobot@mrrobot-virtual-machine:/var/log/snort$

```

```
mrrobot@mrrobot-virtual-machine:/var/log/snort$ dig www.instagram.com

; <<>> DiG 9.18.39-0ubuntu0.22.04.4-Ubuntu <<>> www.instagram.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 45836
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4000
;; QUESTION SECTION:
;www.instagram.com.          IN      A

;; ANSWER SECTION:
www.instagram.com.          1575    IN      CNAME   z-p42-instagram.c10r.instagram.com.
z-p42-instagram.c10r.instagram.com. 6 IN A    57.144.80.34

;; Query time: 1 msec
;; SERVER: 192.168.72.20#53(192.168.72.20) (UDP)
;; WHEN: Tue Jun 09 17:25:57 IST 2026
;; MSG SIZE rcvd: 97
```

```
mrrobot@mrrobot-virtual-machine:/var/log/snort$ cat snort.alert.fast | grep "Instagram"
mrrobot@mrrobot-virtual-machine:/var/log/snort$ cat snort.alert.fast | grep "Insta"
06/09-17:25:57.592519  [**] [1:1000008:0] Insta accessed [**] [Priority: 0] {UDP} 192.168.72.20:53 -> 192.168.74.201:57070
mrrobot@mrrobot-virtual-machine:/var/log/snort$
```